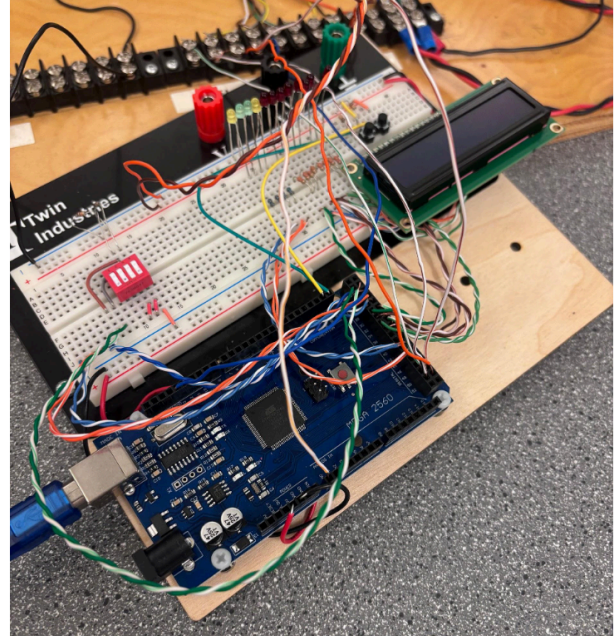
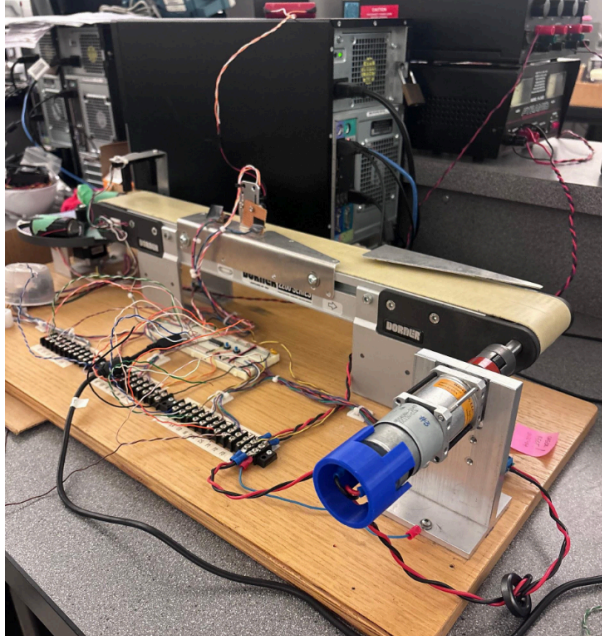


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Automated Sorting Machine



What

- 4th year Mechatronics project.
- Wire and code an ATmega 2560 microcontroller to identify and sort 48 objects based on material and colour.
- Integrate microcontroller with a DC motor, stepper motor, reflectivity sensor, optical sensors, and LCD screen.

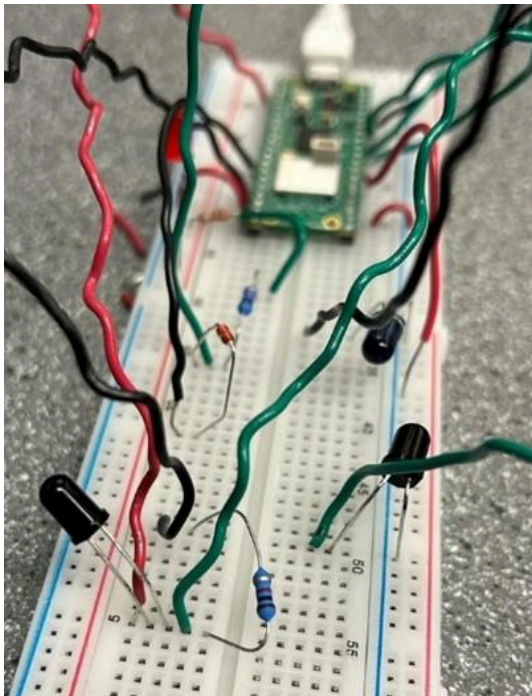
How

- A high level plan was made to outline the needed processes.
- Each physical component was wired and tested.
- Algorithms were developed and tested for each process, like controlling stepper motor speed, scanning objects.
- The algorithms were integrated together optimized for speed.

Results

- The system was able to successfully sort 48 items within 30 seconds, scoring within the top 25% of the class.

Wi-Fi Capable Smart Fire Alarm



Sensor Status

Smoke Detector	☐ ON	☒ Smoke Detector ON	☐ Smoke Detector OFF
Infrared Sensor	☐ ON	☒ Infrared Sensor ON	☐ Infrared Sensor OFF
Temperature Sensor	☐ ON	☒ Temperature ON	☐ Temperature OFF

Fire Status

Smoke Detector LED Status: **Off**
Infrared LED Status: **On**
Temperature LED Status: **Off**

What

- Wire and code a Raspberry Pi Pico microcontroller to detect a possible fire.
- Integrate microcontroller with a smartphone app through Wi-Fi, to display information and control sensors.

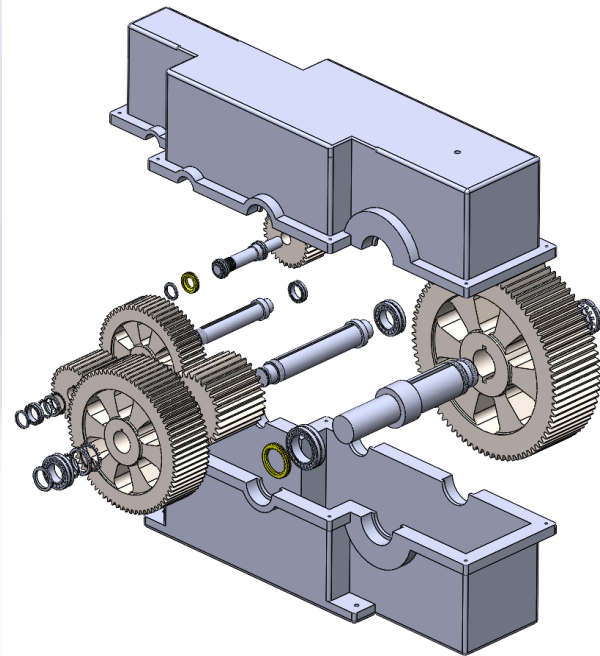
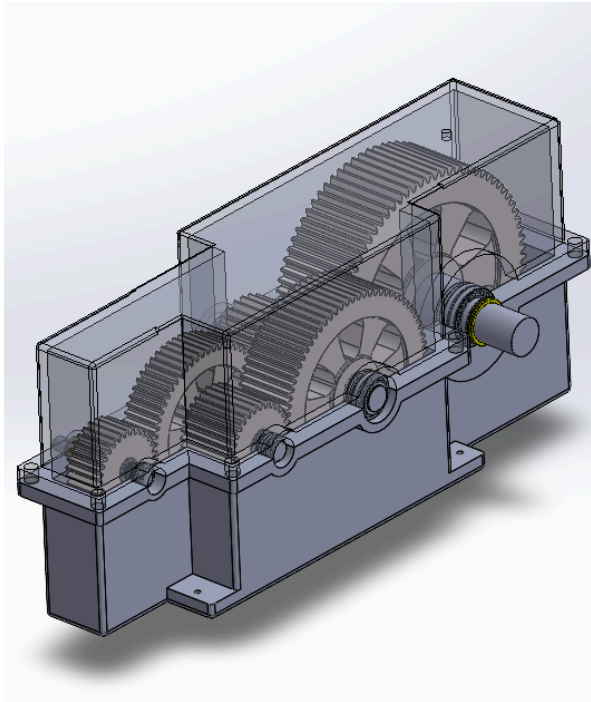
How

- A combination of an infrared sensor, a photoresistor, and a thermistor were chosen to detect fires in three different ways.
- Each physical component was wired and tested.
- A simple website was coded to display the current status of each sensor, and to alert the user if any values were outside of their normal range.
- LEDs were connected to the system to mimic a fire for testing, and baseline sensor values were calibrated.

Results

- The system successfully sent sensor data to the mobile page, proving to be a successful proof of concept.

Steam Turbine Powerplant Gearbox



What

-3th year Mechanical Design project.

-Design all components of a gearbox to be used in a solar powered steam turbine powerplant capable of transmitting 15 MW of power for 25 years.

How

-Research was conducted to analyze designs already on the market.

-The number of gears and speed ratios were decided, and loads/stresses were calculated.

-Components for bearings, couplings, etc. were selected from manufacturer catalogues.

-Recommended maintenance schedule was created.

-The final gearbox housing and layout was modelled and assembled in SolidWorks.

Results

-The final design was able to meet all its requirements and objectives, and received an A+ grade in the class.